

Supporting Information

Ultrahigh-throughput activity engineering of promiscuous amidases through a fluorescence- activated cell sorting assay

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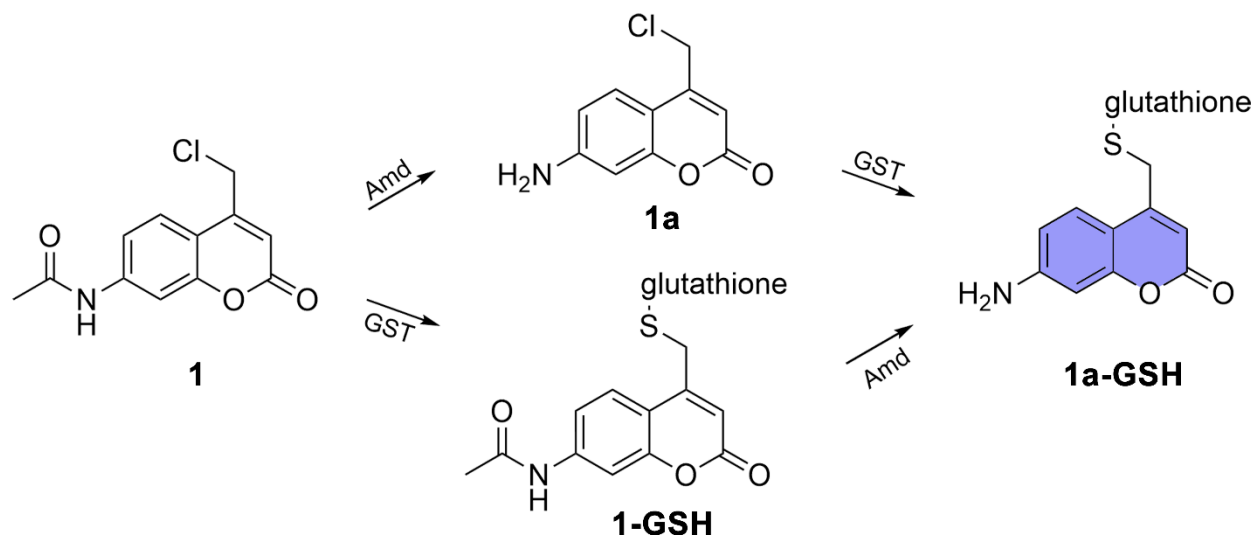
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Supplementary Schemes, Figures and Tables



Scheme S1. The hydrolysis of 7-acetamido-4-chloromethylcoumarin (**1**) by an amidase (Amd) yields 7-amino-4-chloromethylcoumarin (**1a**) and subsequent glutathione *S*-transferase- (GST) catalyzed conjugation to glutathione (**1a-GSH**). Alternatively, **1** is conjugated to **1-GSH** followed by amide hydrolysis. This reaction may occur in both orders, yielding the same fluorescent product.

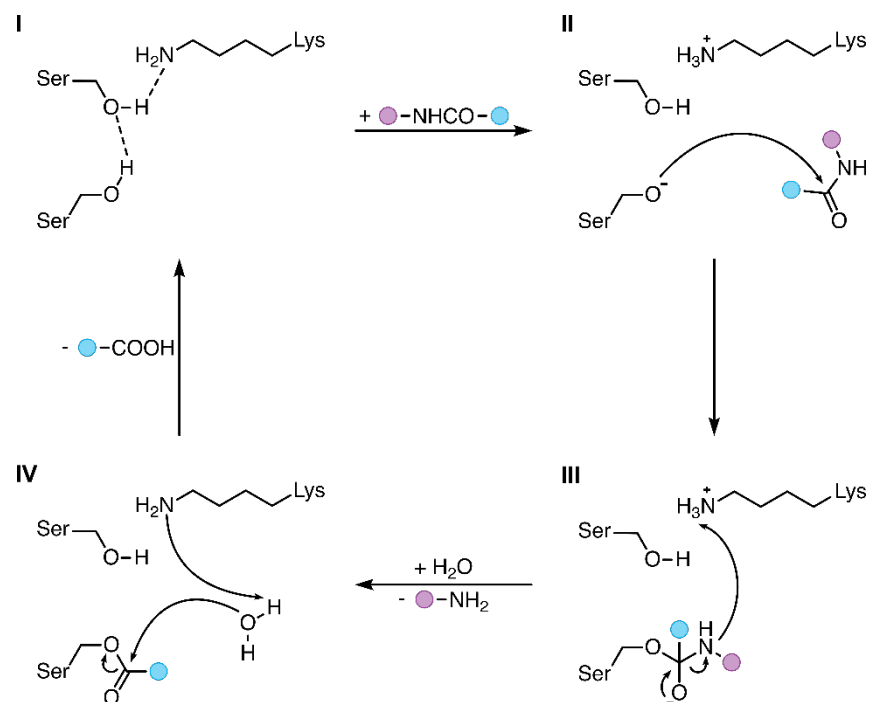


Figure S1. Simplified catalytic mechanism of AS family amidases. **(I)**, The formation of a hydrogen bond network between the catalytic triad residues (Lys-*cis*Ser-Ser) yields a polarized serine. **(II)**, This serine can then perform a nucleophilic attack on the carbonyl carbon in the amide substrate. **(III)**, Formation of the tetrahedral transition state is followed by the release of the amine product. **(IV)**, Finally, the acyl-enzyme intermediate is hydrolyzed by an activated water molecule to liberate the carboxylic acid product. The free enzyme is then reconstituted, completing the catalytic cycle.¹⁻³

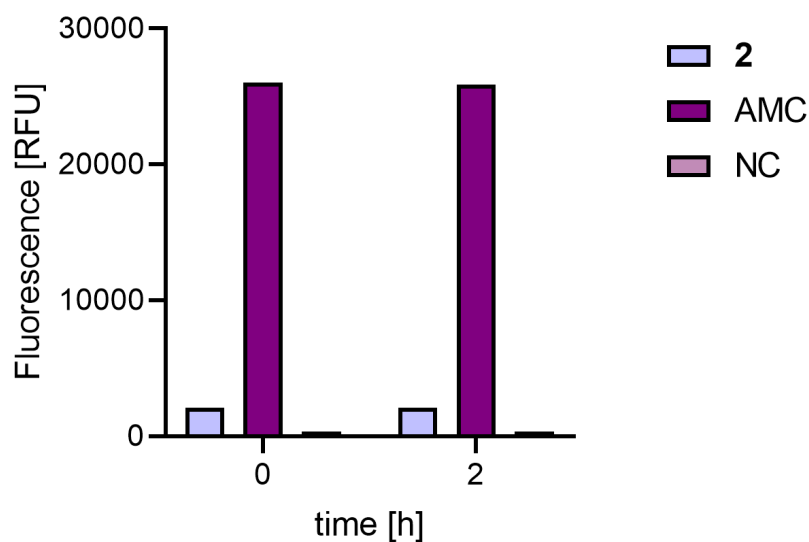


Figure S2. Host background control of *E. coli* BL21(DE3)-Gold cells. Fluorescence signals of *E. coli* BL21(DE3)-Gold cells after addition of 100 μ M substrate **2**, acetamido-4-methylcoumarin (AMC) in 10% (v/v) dimethyl sulfoxide (DMSO; pH 7.5) in LB medium after 0 and 2 hours. Cells supplemented with 10% (v/v) DMSO were used as a negative control (NC). Fluorescence was measured in a 96-well microtiter plate reader ($\lambda_{\text{ex}} = 345$ nm and $\lambda_{\text{em}} = 445$ nm).

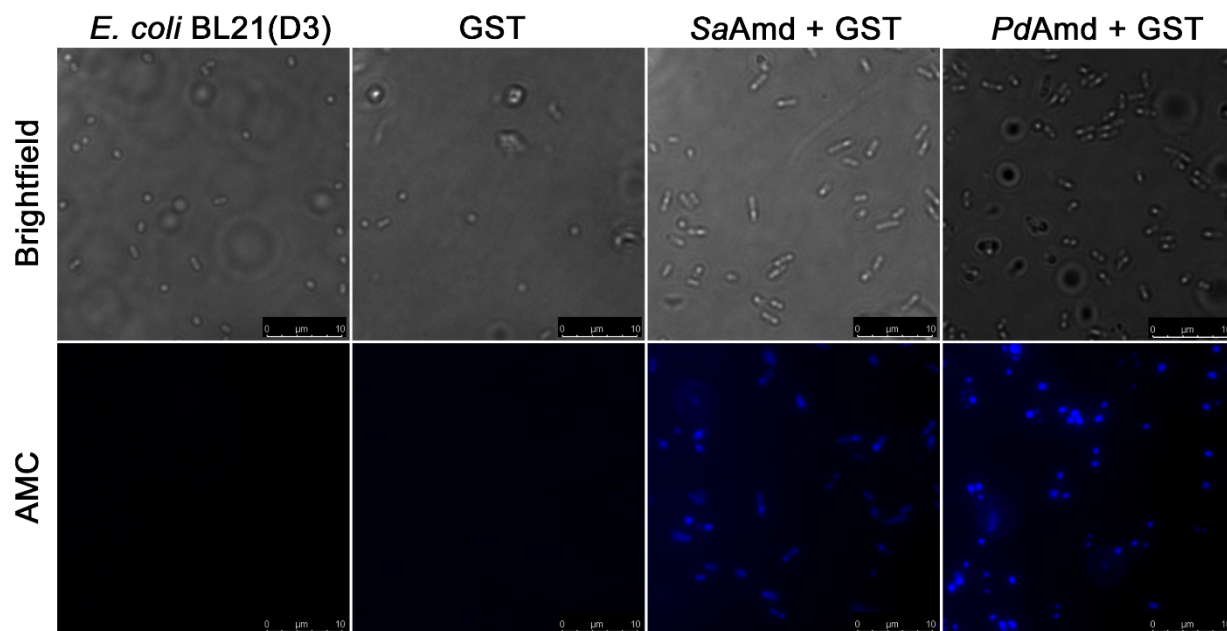


Figure S3. Visualization of the retention of reaction product **1a-GSH** in *E. coli* BL21(DE3)-Gold cells. Prior to flow-cytometry analysis *E. coli* BL21(DE3)-Gold cells without a plasmid, expressing only GST as a control, and co-expressing GST and the amidases *SaAmd* or *PdAmd* respectively, were incubated with **1** (final concentration: 100 μ M) for 24 h during cultivation. The cells were visualized in brightfield and fluorescence detection (DAPI channel, $\lambda_{\text{ex}} = 350/50$ nm and $\lambda_{\text{em}} = 460/50$ nm). Brightfield images show cells present, whereas fluorescence can only be detected for cells expressing *SaAmd* or *PdAmd* together with GST. Scale bar: 10 μ m.

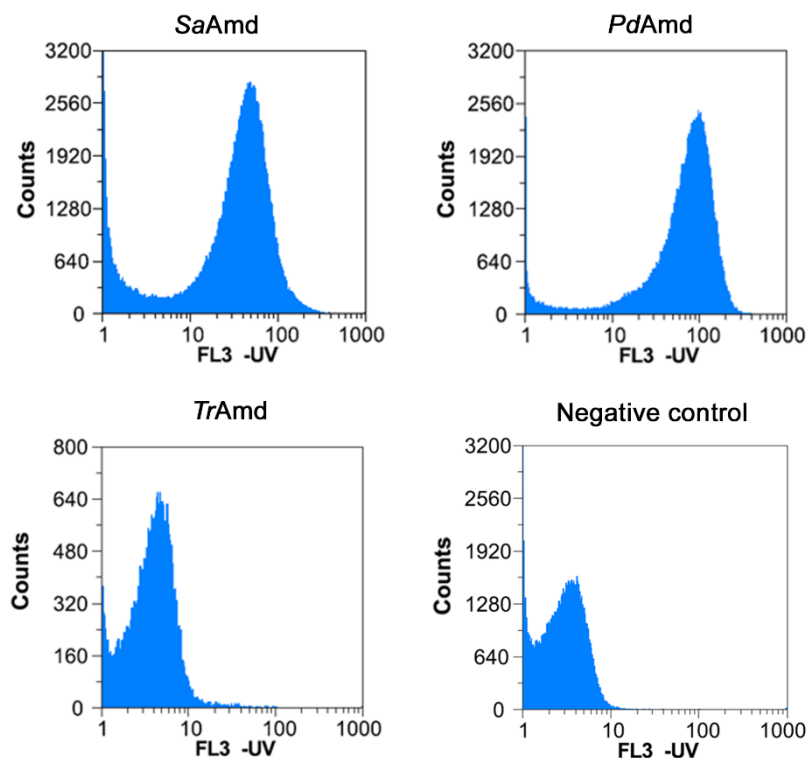


Figure S4. Flow-cytometric analysis of an in-house amidase panel. Fluorescence histograms of *E. coli* BL21(DE3)-Gold cells, co-expressing GST and the amidases *SaAmd*, *PdAmd*, or an amidase from *Thermomicrobium roseum* (*TrAmd*, UniProtKB accession no.: B9L3U7) *E. coli* cells expressing only GST were used as a negative control. Amidase activity in whole cells towards **1** was measured by flow cytometry; (FL3-UV: $\lambda_{\text{ex}} = 375$ nm and $\lambda_{\text{em}} = 440$ nm).

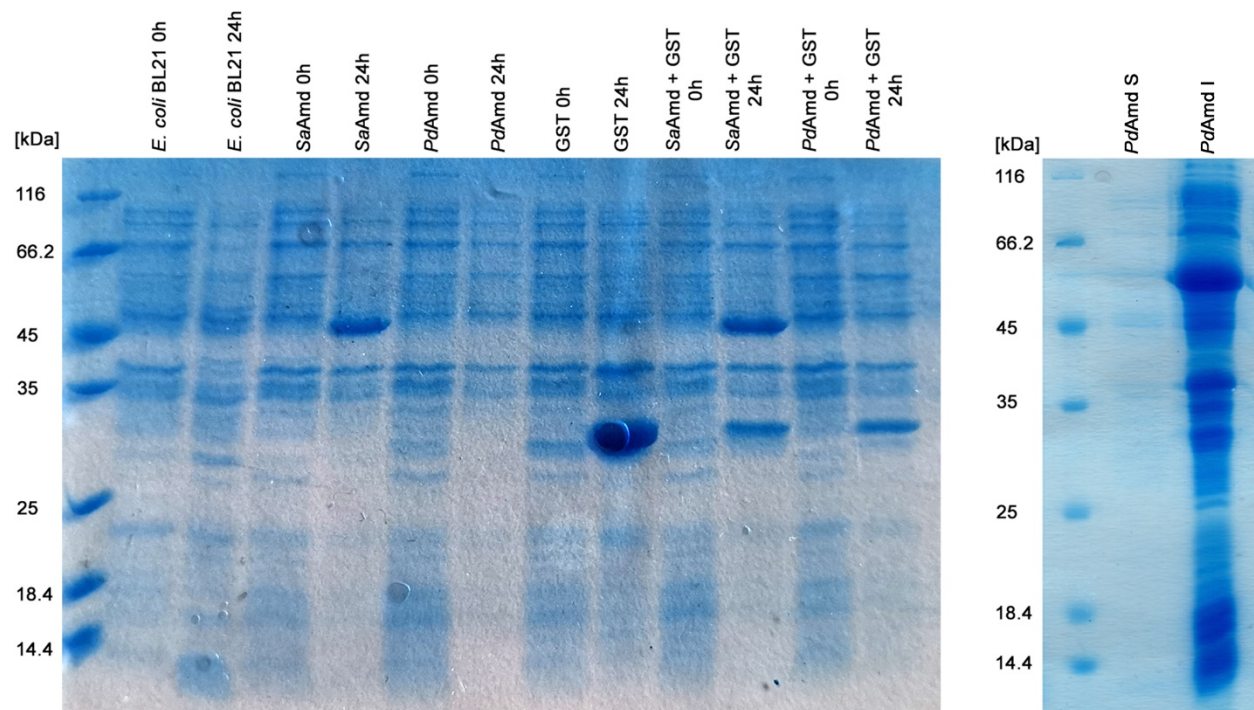


Figure S5. SDS-PAGE analysis of crude cell lysates. 5 μ L of denatured lysate from *E. coli* BL21(DE3)-Gold cells expressing *SaAmd* (~48 kDa), *PdAmd* (~55.5 kDa) and GST (~25.4 kDa), individually and co-expressed, were applied 0 h and 24 h after induction. *E. coli* BL21(DE3)-Gold cells were used as a control. *PdAmd* was separated into soluble (*PdAmd* S) and insoluble (*PdAmd* I) fraction to show successful expression. A marker was used to determine the molecular weight.

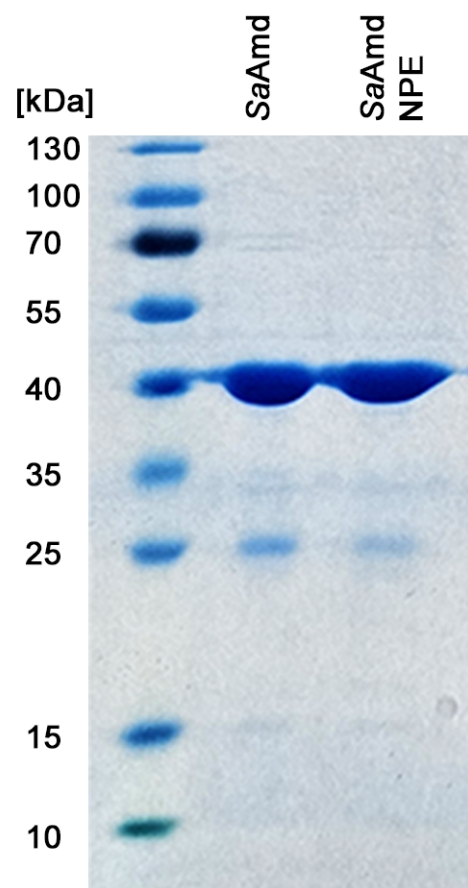


Figure S6. SDS-PAGE of purified *SaAmd*. 5 μ L of purified *SaAmd* and *SaAmd*-NPE representative for all variants (~48 kDa) were applied. A marker was used to determine the molecular weight.

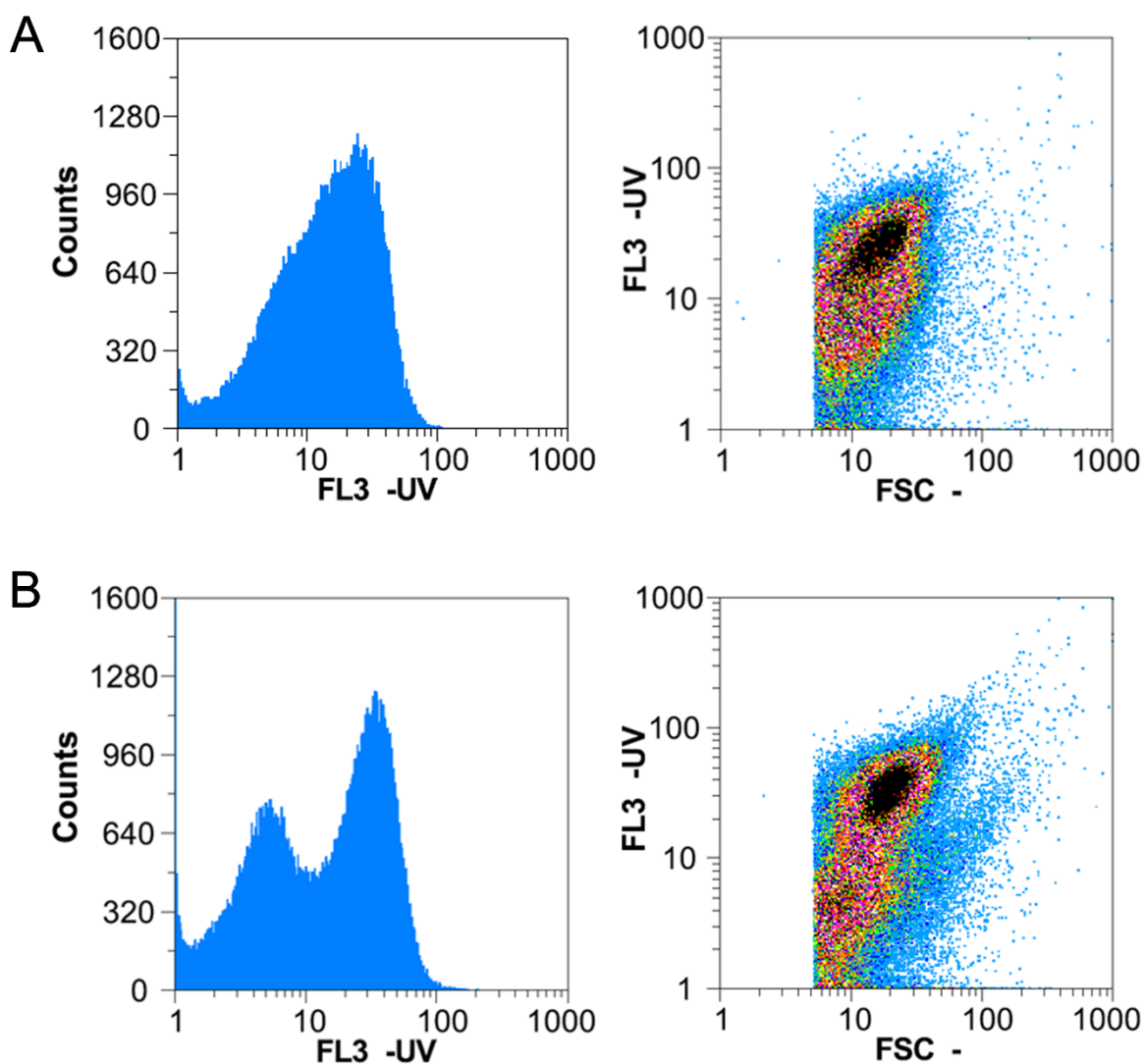


Figure S7. Activity-based enrichment of a cell population by FACS. Fluorescence histograms of a 1:1 mixed population of *E. coli* BL21(DE3)-Gold cells co-expressing GST and *SaAmd* or *PdAmd* (**A**) before and (**B**) after FACS, enrichment, and re-sorting. Amidase activity in whole cells towards **1** was measured by flow cytometry; (FL3-UV: $\lambda_{\text{ex}} = 375$ nm and $\lambda_{\text{em}} = 440$ nm), FSC: forward scatter.

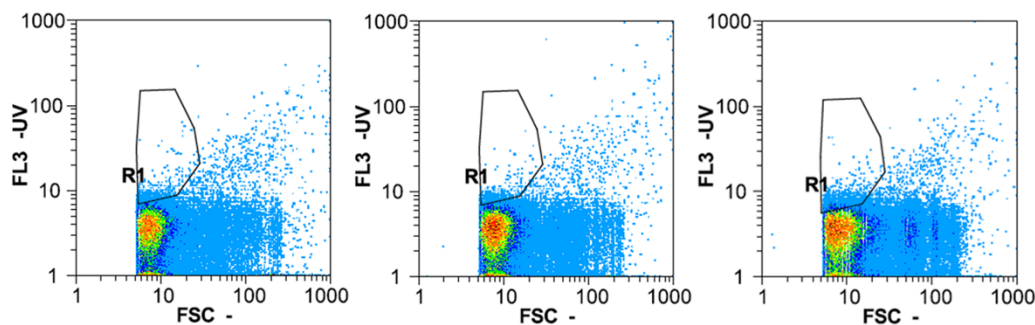


Figure S8. Representative sorting campaigns of *SaAmd* libraries A207NNK/R208NNK, R208NNK/S209NNK and S209NNK/V210NNK. The top 0.5–1% of fluorescent cells were sorted (Gate R1). (FL3-UV: $\lambda_{\text{ex}} = 375$ nm and $\lambda_{\text{em}} = 440$ nm), FSC: forward scatter.

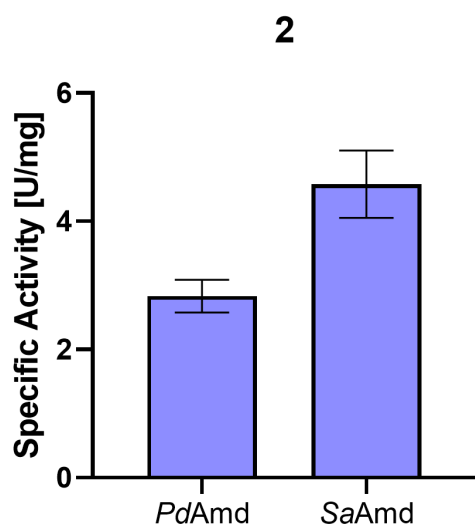


Figure S9. Specific activities of *PdAmd* and *SaAmd* on **2**. Hydrolysis of 100 μ M substrate in 50 mM Tris-HCl, 100 mM NaCl, 10% (v/v) dimethyl sulfoxide (DMSO; pH 7.5) at 22–24 °C. Recording of the increase in fluorescence ($\lambda_{\text{ex}} = 365$ nm and $\lambda_{\text{em}} = 440$ nm) from the formation of the corresponding hydrolysis products over time (0–3 min). Specific activities are shown as mean values $\pm$ standard deviation (SD) of biological replicates ($n \geq 3$).

Table S1. Primers used in this study.

Position	Primer	Primer sequence (5' – 3')
<i>SaAmd G62</i>	fwd	cgccgtctggcagccnnkgataccgcaccgctg
	rev	cagcgggtgcggatcmnnggctgccagacggcg
<i>SaAmd D63</i>	fwd	tcgccgtctggcagccggtnnkaccgcaccgc
	rev	gcggtgcggtmnnaccggctgccagacggcga
<i>SaAmd T64</i>	fwd	tctggcagccggtgatnnkgcaccgctgctggg
	rev	cccagcagcgggtgcmnnatcaccggctgccaga
<i>SaAmd T86</i>	fwd	gccggcctgccgaccnnktggggcttgaagcc
	rev	ggcttgaagcccamnnggtcggcaggccggc
<i>SaAmd T87</i>	fwd	ggcctgccgaccacnnkggcttgaagcccat
	rev	atgggcttgaagccmnnggtggtcggcaggcc
<i>SaAmd V122</i>	fwd	ggtaaaaccaatgtgccgnnkagcctggccgatctgcag
	rev	ctgcagatcggccaggctmnnccggcacattggtttacc
<i>SaAmd S123</i>	fwd	taaaaccaatgtgccggtgnnkctggccgatctgcagagt
	rev	cactctgcagatcggccagmnnccaccggcacattggttta
<i>SaAmd V147</i>	fwd	catgatctgagtcgtgtgnnkgcggtagtagtggcggt
	rev	accgccactactaccgcmnnccacagactcagatcatg
<i>SaAmd P148</i>	fwd	atctgagtcgtgtgccgnnkggtagtagtggcggtgc
	rev	gcaccgccactactaccmnnccggcacacgactcagat
<i>SaAmd D170</i>	fwd	gccgctggaatatggcagtnnkattggcggtagcattcgcg
	rev	cgcgaatgctaccgccaatmnnactgccatattccagcggc
<i>SaAmd I171</i>	fwd	tgccgctggaatatggcagtgatnnkggcggtagcattcg
	rev	cgaatgctaccgcmnnatcactgccatattccagcggca
<i>SaAmd G192</i>	fwd	tggggtcataaaagtaccttcnnkgtgctgccgaatgaaggcac
	rev	gtgaccttcattcggcagcacmnnngaagggtactttatgaccca

<i>SaAmd V193</i>	fwd	ggtcataaaagtaccttcggcnnkctgccgaatgaaggctcacttc
	rev	gaagtgaccttcattcggcagmnngccgaaggctctttatgacc
<i>SaAmd L194</i>	fwd	cataaaagtaccttcggcgtgnnkccgaatgaaggctcacttcttc
	rev	gaagaagtgaccttcattcggmnncacgccgaaggctctttatg
<i>SaAmd P195</i>	fwd	aaaagtaccttcggcgtgctgnnkaatgaaggctcacttcttcccg
	rev	cgggaagaagtgaccttcattmnncagcacgccgaaggctctttt
<i>SaAmd N196</i>	fwd	ccttcggcgtgctgccgnnkgaggctcacttcttccc
	rev	gggaagaagtgaccttcmnncggcagcacgccgaagg
<i>SaAmd G206</i>	fwd	cttcttccgggcaccgatnnkgcacgtagtgttctggc
	rev	gccagaacactacgtgcmnnatcgggtcccgggaagaag
<i>SaAmd A207</i>	fwd	ttcccgggcaccgatggtnnkcgtagtgttctggcc
	rev	ggccagaacactacgmnnaccatcgggtcccgggaa
<i>SaAmd R208</i>	fwd	cgggcaccgatggtgcannkagtgttctggccgtg
	rev	cacggccagaacactmnntgcaccatcgggtcccg
<i>SaAmd S209</i>	fwd	gcaccgatggtgcacgtnnkgttctggccgtgattgg
	rev	ccaatcacggccagaacmnnacgtgcaccatcgggtgc
<i>SaAmd V210</i>	fwd	ccgatggtgcacgtagnnkctggccgtgattggtcc
	rev	ggaccaatcacggccagmnnectacgtgcaccatcgg
<i>SaAmd G246</i>	fwd	gcaattgaaagtccgcgcnnkctgcgcattctgctgctg
	rev	cagcagcagaatgcgcagmnngcgcggactttcaattgc
<i>SaAmd L247</i>	fwd	ttgaaagtccgcgcggcnnkcgcatctgctgctgac
	rev	gtcagcagcagaatgcgmnnngcgcgcggacttcaa
<i>SaAmd H255</i>	fwd	cattctgctgctgaccgaannkccgagcgccaaagtgg
	rev	ccactttggcgctcggmnnttcggtcagcagcagaatg
<i>SaAmd P256</i>	fwd	gcgcattctgctgctgaccgaacatnnkagcgccaaagtgg
	rev	ccactttggcgctmnnatgttcggtcagcagcagaatgcgc

Gibson assembly of pACYCDuet-1_gst		
GST_insert	fwd	cccctgtagaaataatTTTgtttaactttaataaggagatatacc atgaccaagttacctatactaggttattggaaaattaagg
	rev	cgtgtacaatacgaattactttctgttcgacttaagtcaatgacgc ggaaccagatgatcc
pACYC_backbone	fwd	ggatcatctggtccgcgtcattgacttaagtcgaacagaaag taatcgattgtacacg
	rev	aatgcaccggactatgacttttaccattgtatatctccttctat acttaactaatatactaagatggggaattg
Sequencing Primer		
pET26b(+)_fwd	fwd	gtagaggatcgagatctcgatcccgc
SaAmd_reverse	rev	ctcgaggccaattgcacat

Table S2. Libraries screened in this work.

SaAmd libraries
G62NNK D63NNK
D63NNK T64NNK
T86NNK T87NNK
V122NNK S123NNK
V147NNK P148NNK
D170NNK I171NNK
G192NNK V193NNK
V193NNK L194NNK
L194NNK P195NNK
P195NNK N196NNK
G206NNK A207NNK
A207NNK R208NNK
R208NNK S209NNK
S209NNK V210NNK
G246NNK L247NNK
H255NNK P256NNK
S123NNK R208NNK A364NNK

Table S3. Specific activities of SaAmd and variants towards substrates **1–8**.

Substrate / Variant	1-GSH [U/mg]	2 [U/mg]	3 [U/mg]	4 [U/mg]	5 [U/mg]	6 [U/mg]	7 [U/mg]	8 [U/mg]
SaAmd WT	0.30±0.06	4.58±0.48	7.53±0.60	3.65±0.67	1.10±0.37	3.66±0.34	0.69±0.21	0.07±0.02
A207S	0.26±0.01	5.14±0.64	13.90±0.44	3.16±0.39	0.96±0.01	10.40±0.93	0.53±0.02	0.05±0.01
R208S	0.14±0	1.12±0.07	2.10±0.54	1.72±0.36	0.33±0.02	2.68±0.08	0.22±0.05	0.03±0
R208Q	0.19±0.01	5.77±0.63	8.54±0.77	3.29±0.32	0.83±0.02	7.80±0.41	0.57±0.04	0.06±0.004
R208E	0.08±0	6.84±0.38	12.22±3.11	2.94±0.55	0.84±0.07	9.80±3.77	0.81±0.03	0.02±0.002
R208P	0.25±0.02	3.86±0.73	8.74±0.27	2.15±0.19	0.74±0.03	6.58±2.06	0.36±0.01	0.05±0.002
S209P	0.14±0	5.29±0.59	9.92±1.4	5.87±0.38	1.93±0.08	6.43±0.97	0.45±0.07	0.19±0.01
S209Q	0.06±0	1.99±0.38	5.79±0.64	2.31±0.26	0.58±0.02	3.76±0.62	0.37±0.01	0.04±0.003
V210P	0.10±0.01	6.49±0.21	12.32±0.55	6.16±0.65	1.31±0.07	12.92±0.48	0.72±0.02	0.09±0.006
S123A	0.42±0.01	7.49±0.39	15.99±0.47	4.62±0.9	1.23±0.07	8.86±0.76	0.91±0.11	0.10±0.002
S123N	0.31±0.04	7.99±0.51	7.02±1.2	10.07±0.53	1.13±0.02	6.99±0.74	0.95±0.17	0.11±0.003
A364P	0.35±0.04	7.00±0.27	15.80±0.12	4.98±0.69	1.62±0.17	8.07±0.4	0.69±0.06	0.10±0.004
A364E	0.31±0.04	5.07±0.34	13.26±2.51	4.71±0.38	1.18±0.06	7.75±0.49	0.62±0.03	0.08±0.001
A364S	0.18±0.01	1.98±0.45	4.34±0.49	1.48±0.26	0.56±0.03	1.48±0.57	0.34±0.02	0.05±0.001
A207S R208S	0.58±0.13	1.72±0.12	4.52±0.47	5.08±0.41	1.70±0.17	16.42±0.74	1.23±0.04	0.09±0
R208S S209Q	0.07±0.04	1.31±0.03	2.74±0.53	9.33±1.46	1.56±0.18	20.34±4.18	1.35±0.11	0.09±0.01
R208Q S209P	0.17±0.01	3.88±0.11	10±0.35	11.02±1.67	2.69±1.60	6.05±2.86	1.18±0.01	0.27±0.01
S209P V210P	0.35±0.02	6.04±0.46	13.65±0.73	17.15±0.82	3.43±2.37	7.28±3.67	1.59±0.06	0.42±0.03
S123A R208E A364S	0.13±0.01	27.92±1.48	47.44±0.89	18.18±2.21	1.79±0	29.73±6.01	2.64±0.14	0.11±0.04
S123N R208P A364E	1.12±0.06	4.08±0.24	14.75±0.19	59.73±1.75	3.36±0.07	45.58±3.09	4.03±0.27	0.31±0.04

R208P A364P	1.33±0.12	2.30±0.30	7.82±1.16	14.84±2.46	3.56±0.21	34.89±10.0 1	2.14±0.19	0.21±0.04
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Table S4. Detection of hydrolysis products and determined molar absorption coefficient.

Substrate	Product	Wavelength	Molar absorption coefficient
1-GSH	1a-GSH	$\lambda_{\text{ex}} = 365 \text{ nm};$ $\lambda_{\text{em}} = 440 \text{ nm}$	6,352
2	AMC	$\lambda_{\text{ex}} = 365 \text{ nm};$ $\lambda_{\text{em}} = 440 \text{ nm}$	61,962
3	AN	$\lambda_{\text{ex}} = 315 \text{ nm};$ $\lambda_{\text{em}} = 445 \text{ nm}$	10,753
4	pNA	$\lambda = 390 \text{ nm}$	4.9992
5	pNA	$\lambda = 390 \text{ nm}$	4.9992
6	UMB	$\lambda_{\text{ex}} = 365 \text{ nm};$ $\lambda_{\text{em}} = 440 \text{ nm}$	61,962
7	AMC	$\lambda_{\text{ex}} = 365 \text{ nm};$ $\lambda_{\text{em}} = 440 \text{ nm}$	61,962
8	pNA	$\lambda = 390 \text{ nm}$	4.9992

Cloning protocol

pACYCDuet-1_gst assembly

Gibson assembly[®] primers were designed following the instructions of the supplier. The Q5[®] High-Fidelity DNA Polymerase was used to amplify the pACYCDuet-1 backbone and the GST insert following the instructions of the supplier. The initial denaturation was performed at 98 °C

for 30 s. This was followed by 30 cycles of denaturation at 98 °C for 5 s, annealing at 60 °C for 30 s and elongation at 72 °C for 30 s/kb. The final extension was performed for 2 min at 72 °C. The successful amplification of the fragments was confirmed on an agarose gel. DpnI digestion was performed for 2 h at 37 °C. The NucleoSpin™ Gel and PCR Clean-up Kit (Macherey-Nagel GmbH & Co. KG, Düren, Germany) was used to purify DNA fragments. The Gibson assembly was achieved using Gibson Assembly® Master Mix from New England Biolabs® and following the instructions of the supplier. The Gibson assembly product was transformed into *E. coli* TOP10 following standard transformation protocol. Colonies were picked from an LB agar plate (supplemented with 25 µg·mL⁻¹ chloramphenicol) the following day and used to inoculate an overnight culture (5 mL, 25 µg·mL⁻¹ chloramphenicol, 37°C, 140 rpm). The plasmids were isolated and successful construction of pACYCDuet-1_ *gst* was confirmed by Sanger sequencing.

Chemical syntheses of 1-GSH and 1a-GSH

NMR-spectroscopy was performed using an Avance II 300 (Bruker Daltonics GmbH & Co. KG, Bremen, Germany, ¹H: 300 MHz, ¹³C: 75 MHz) and a Bruker Avance IIIHD 600 spectrometer equipped with a Prodigy BBO cryo probe (¹H: 600 MHz, ¹³C: 151MHz). Chemical shifts are given in parts per million (ppm) and were calibrated with internal standard of deuterium-labeled dimethyl sulfoxide (DMSO, ¹H 2.50 ppm, ¹³C 39.52 ppm). NMR assignments of new compounds were confirmed by ¹H - ¹H COSY, ¹H - ¹H, ¹H - ¹³C, HSQC, and ¹H - ¹³C, HMBC. Proton multiplicities are denoted by the following abbreviations: s (singlet), d (doublet), dd (doublet of doublets), ddd, (doublet of doublet of doublets), dh (doublet of heptets) t (triplet), td (triplet of doublets), q (quartet), m (multiplet). Coupling constants (*J*) are presented in Hertz (Hz).

Thin layer chromatography (TLC) analysis was performed on aluminum-backed unmodified Merck silica gel 60 F245 plates. Visualization was realized under UV irradiation.

Reversed-phase high-performance liquid chromatography (RP-HPLC) was carried out with an Autopurification system from Waters using an ACQUITY QDa Detector in combination with a 2998 Photodiode Array Detector. Analytical separation was conducted using XSELECT CSH C18 5 μ m 4.6 x 150 mm columns. Preparative separation was performed using XSELECT CSH Prep C18 5 μ m OBD 30 x 150 mm columns. Milli-Q water containing 0.1 vol% formic acid was used as Solvent A, and HPLC grade acetonitrile was used as Solvent B.

UHPLC-MS/UV analysis was performed on a Nexera X2 UHPLC system (Shimadzu) comprised of LC-30AD pumps, SIL-30AC autosampler, CTO-20AC column oven, and DGU-20A5/3degasser module. Detection was done using an SPD-M20A photodiode array and an LCMS-2020 mass spectrometer (ESI/APCI). If not stated otherwise, all separations were performed using a Waters XSelect CSH C18 2.5 μ m (3.0 x 50 mm) column XP at 40 °C and a flow rate of 1.7 mL/min. Gradient elution was performed using mixtures of HPLC grade acetonitrile and HPLC grade water with either 0.1 vol% formic acid (acidic separation).

Synthetic Procedures

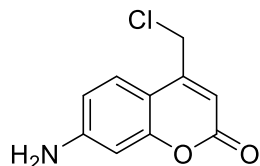
General Operating Procedure A:

Glutathione-coumarin conjugates **1-GSH** and **1a-GSH** were synthesized according to the following modified literature procedure:⁴

A 100 mL round-bottom flask was charged with glutathione (244 mg, 0.8 mmol, 2.0 equiv.) and the solid was dissolved by adding water (25 mL) and acetonitrile (25 mL). The appropriate coumarin derivative (0.4 mmol, 1.0 equiv.) and potassium carbonate (221 mg, 1.6 mmol, 4.0 equiv.) were further added and the resulting suspension was stirred at room temperature overnight. Over time, the reaction mixture turned into a clear solution. The clear solution was further

concentrated under reduced pressure using a rotary evaporator to remove the organic solvent. The remaining aqueous phase was freeze-dried to obtain the crude product, which was subsequently purified by preparative RP-HPLC.

Synthesis of **1a**



1a was prepared from **1**.

1 (4.0 g, 14.2 mmol, 1.0 equiv.) was dissolved in a mixture of 90 mL Tris-HCl (50 mM, 100 mM NaCl, pH 7.5) and 10 mL ethanol. Purified UMG-SP-1⁵ was added to a final concentration of 1 mg mL⁻¹ and the reaction was stirred at room temperature. Throughout the reaction, the pH was maintained at 7.5 using an 877 Titrino plus (Metrohm, Filderstadt, Germany) with 0.5 N NaOH and 0.5 N HCl. After TLC analysis indicated full conversion of the starting material, the reaction was diluted with water (100 mL), and the resulting suspension was adjusted to a slightly basic pH with 1 N aqueous sodium hydroxide solution under cooling in an ice-water bath. The obtained solids were filtered, washed with cold water, and dried under reduced pressure to yield **1a** as a yellow powder (1.7 g, 57%).

¹H NMR (300 MHz, DMSO) δ 7.48 (d, J = 8.7 Hz, 1H), 6.59 (dd, J = 8.7, 2.0 Hz, 1H), 6.44 (s, 1H), 6.18 (s, 2H), 4.87 (s, 2H).

¹³C NMR (75 MHz, DMSO) δ 160.6, 155.9, 153.4, 151.4, 126.1, 111.3, 107.9, 106.1, 98.7, 41.4.

The ¹H-NMR (**Figure S10**) and ¹³C-NMR (**Figure S11**) are in accordance with the literature.⁶

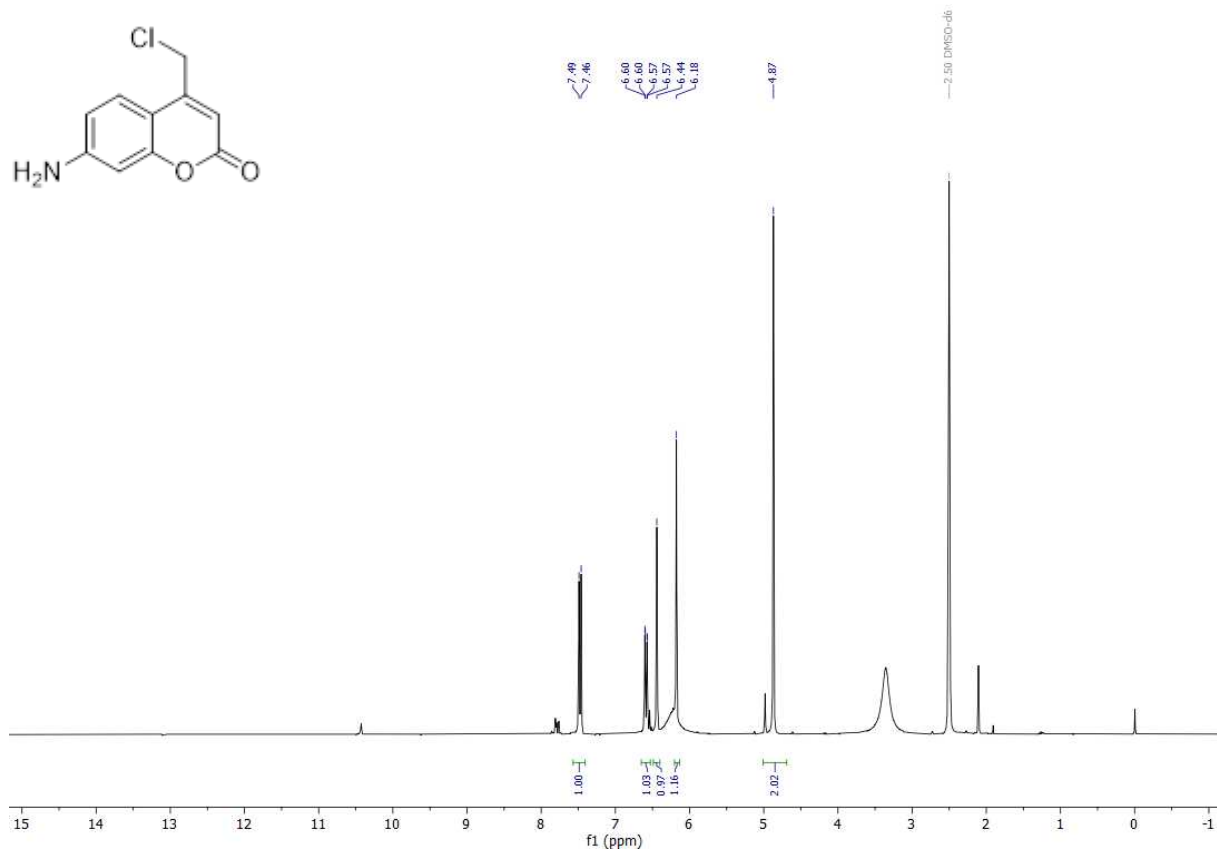


Figure S10. ¹H-NMR of 1a.

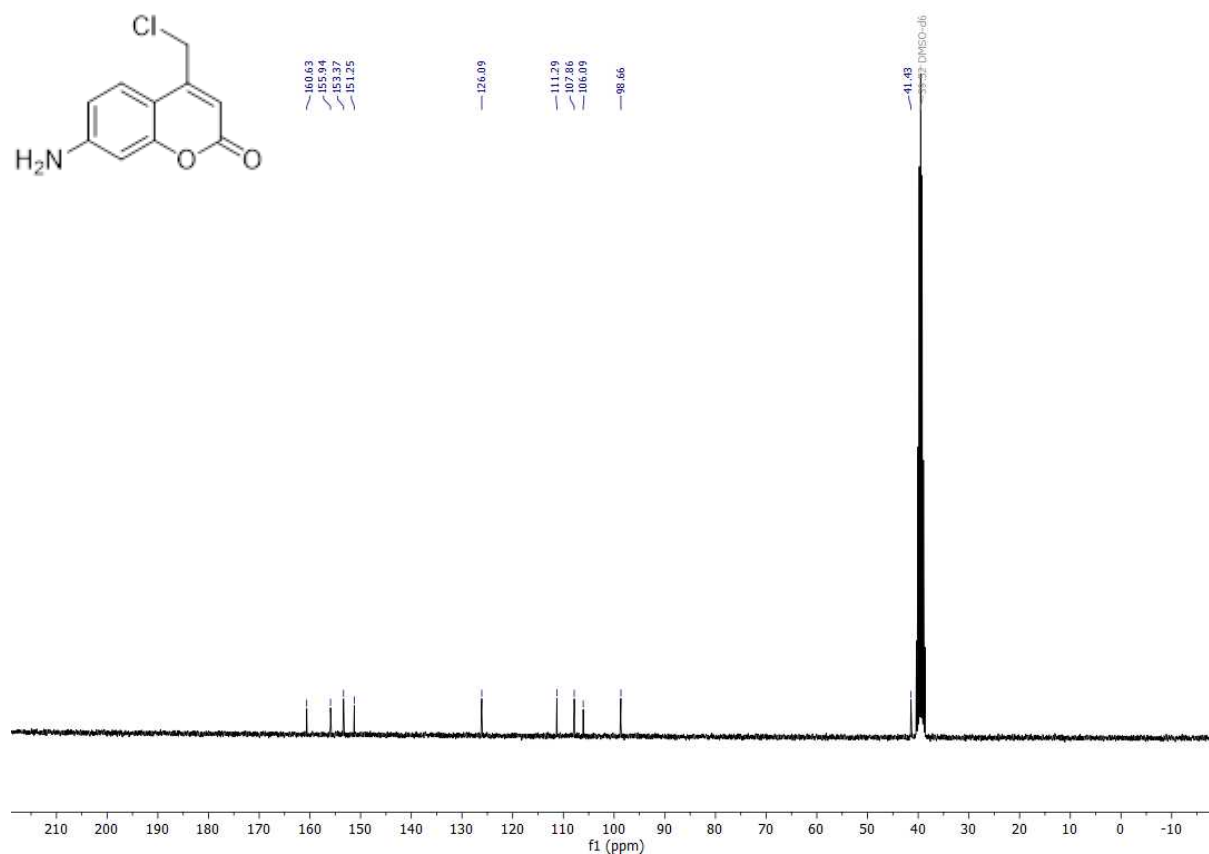
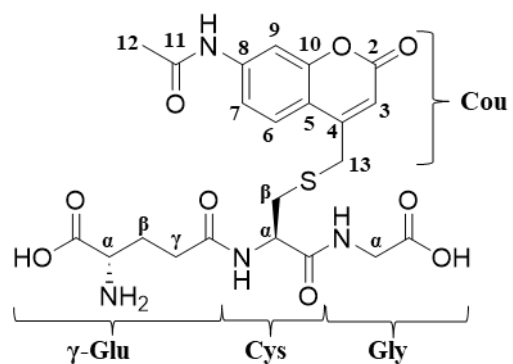


Figure S11. ^{13}C -NMR of **1a**.

Synthesis of **1-GSH**



1-GSH was synthesized according to general procedure A using coumarin derivative **1** (0.4 mmol, 100 mg).

The crude material was purified by preparative RP-HPLC using a gradient of 4%/min (5-95% acetonitrile in MilliQ water with 0.1% formic acid), yielding **1-GSH** as a colorless powder (39 mg,

19%; 523.03 amu). ^1H NMR, ^{13}C NMR, COSY, HMBC and HSQC were measured (**Figure S12-S17**).

^1H NMR (600 MHz, DMSO) δ 10.48 (s, CouHN), 8.70 (t, $J = 5.9$ Hz, GlyHN), 8.41 (d, $J = 8.6$ Hz, CysHN), 7.79 (d, $J = 2.0$ Hz, CouH6), 7.76 (d, $J = 8.8$ Hz, CouH9), 7.40 (dd, $J = 8.8, 2.1$ Hz, CouH7), 6.38 (s, CouH3), 4.53 (td, $J = 9.0, 4.7$ Hz, CysH α), 3.98 – 3.88 (m, CouH13), 3.77 – 3.68 (m, GlyH α), 3.37 (GluH α)*, 2.91 (dd, $J = 13.8, 4.7$ Hz, CysH β 1), 2.65 (dd, $J = 13.7, 9.4$ Hz, CysH β 2), 2.37 – 2.25 (m, GluH γ), 2.10 (s, CouH12), 1.90 (dh, $J = 27.3, 7.1$ Hz, GluH β).

Note: *GluH α overlays with solvent peak. Assigned according to ^1H - ^1H COSY/HSQC.

^{13}C NMR (151 MHz, DMSO) δ 171.8 (GluCOH), 170.9 (CysC=O), 170.6 (GluC=O), 170.2 (GlyC=O), 169.2 (CouC11), 160.0 (CouC2), 154.3 (CouC8), 151.9 (CouC4), 142.6 (CouC10), 126.2 (CouC6), 114.9 (CouC7), 113.0 (CouC5), 112.3 (CouC3), 105.6 (CouC9), 53.0 (GluC α), 52.1 (CysC α), 41.0 (GlyC α), 33.4 (CysC β), 31.4 (GluC γ), 31.1 (CouC13), 26.7 (GluC β), 24.2 (CouH12).

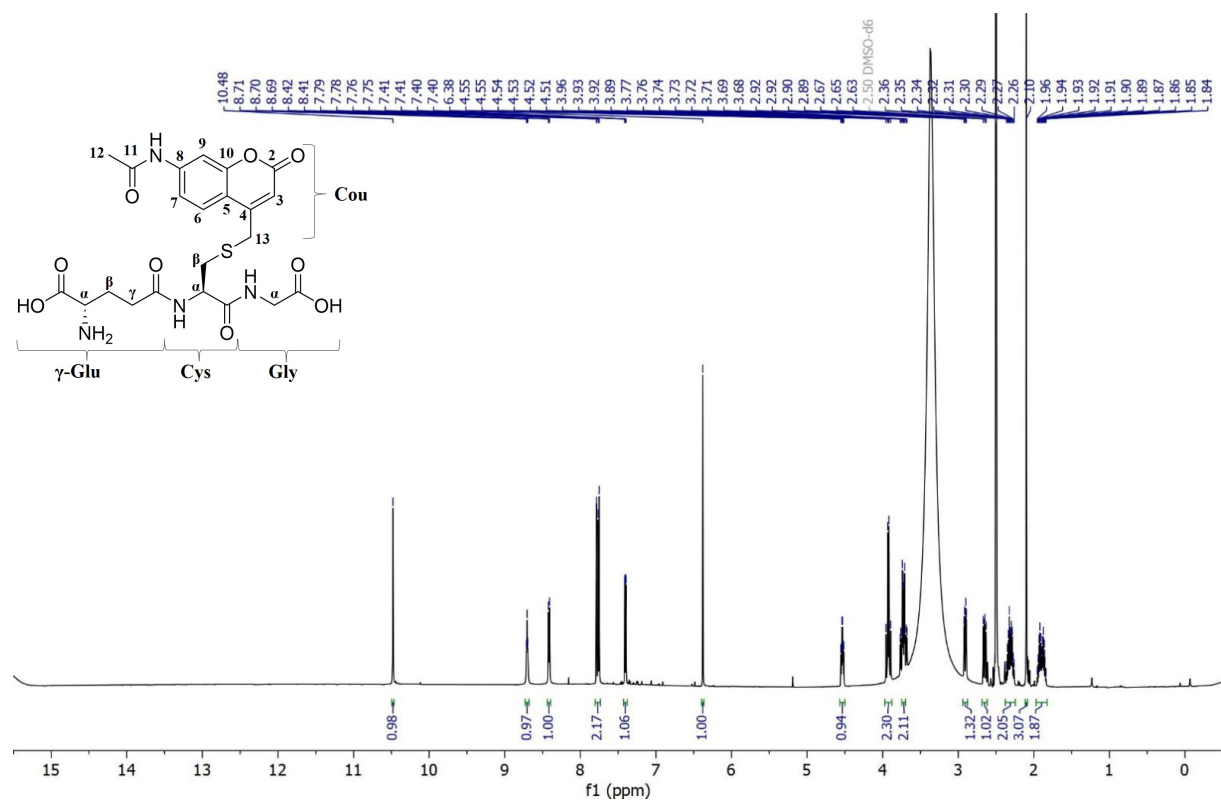


Figure S12. ^1H -NMR of 1-GSH.

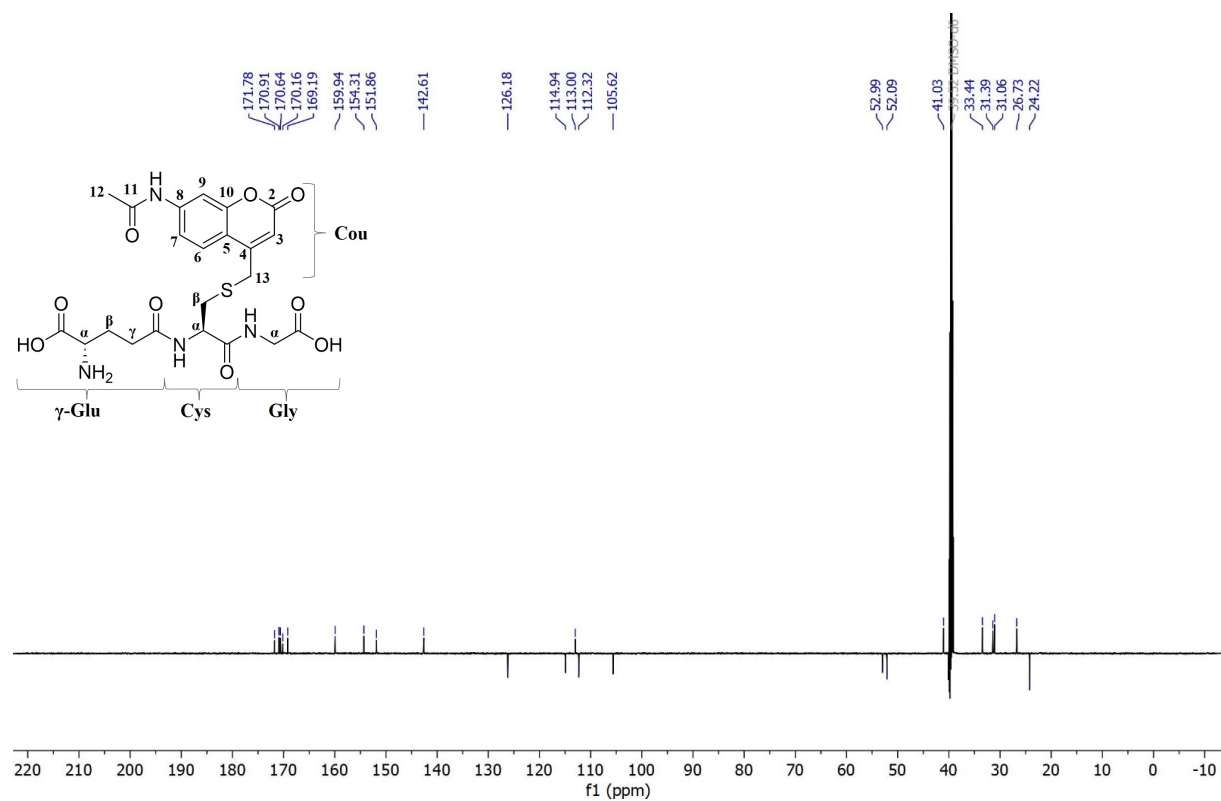


Figure S13. ¹³C-NMR of 1-GSH.

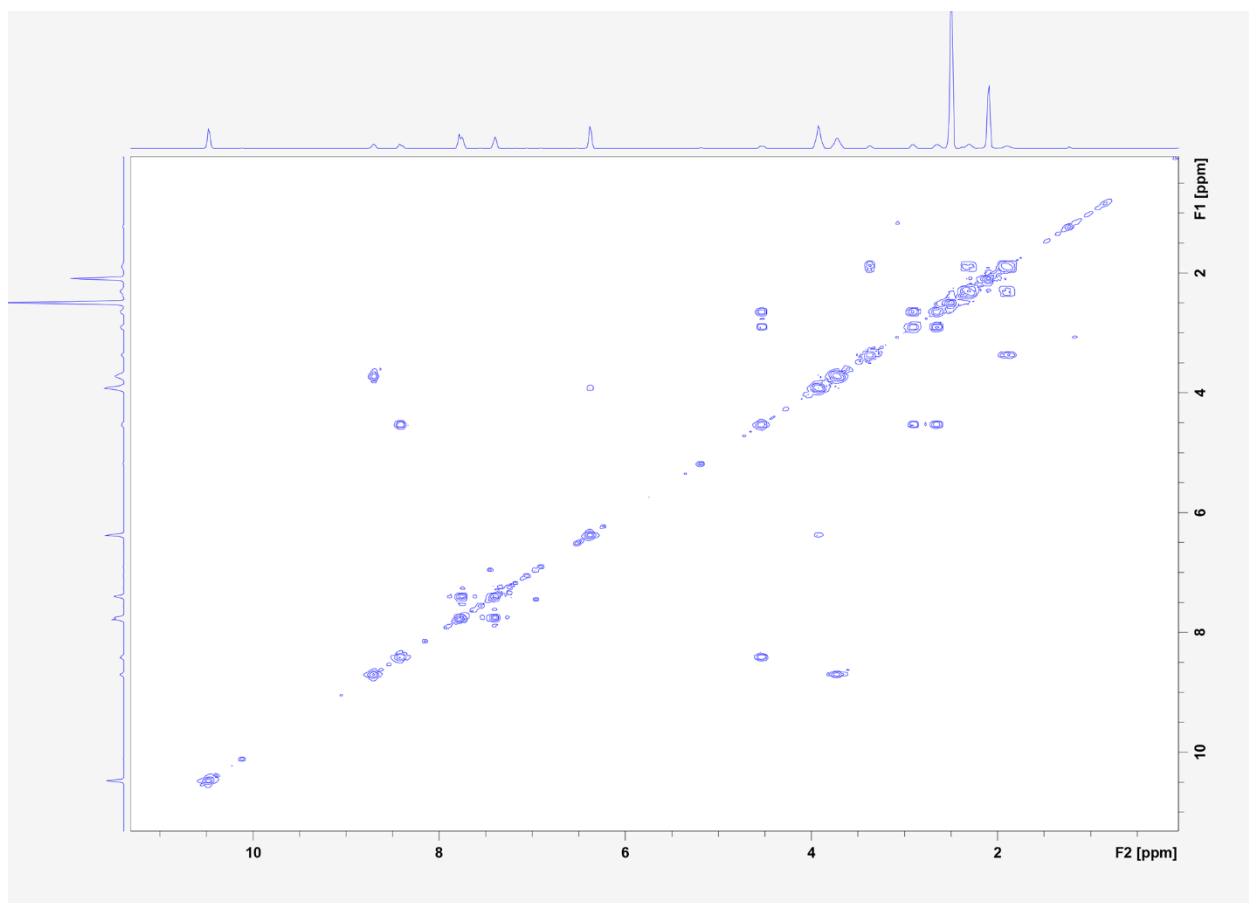


Figure S14. COSY of 1-GSH.

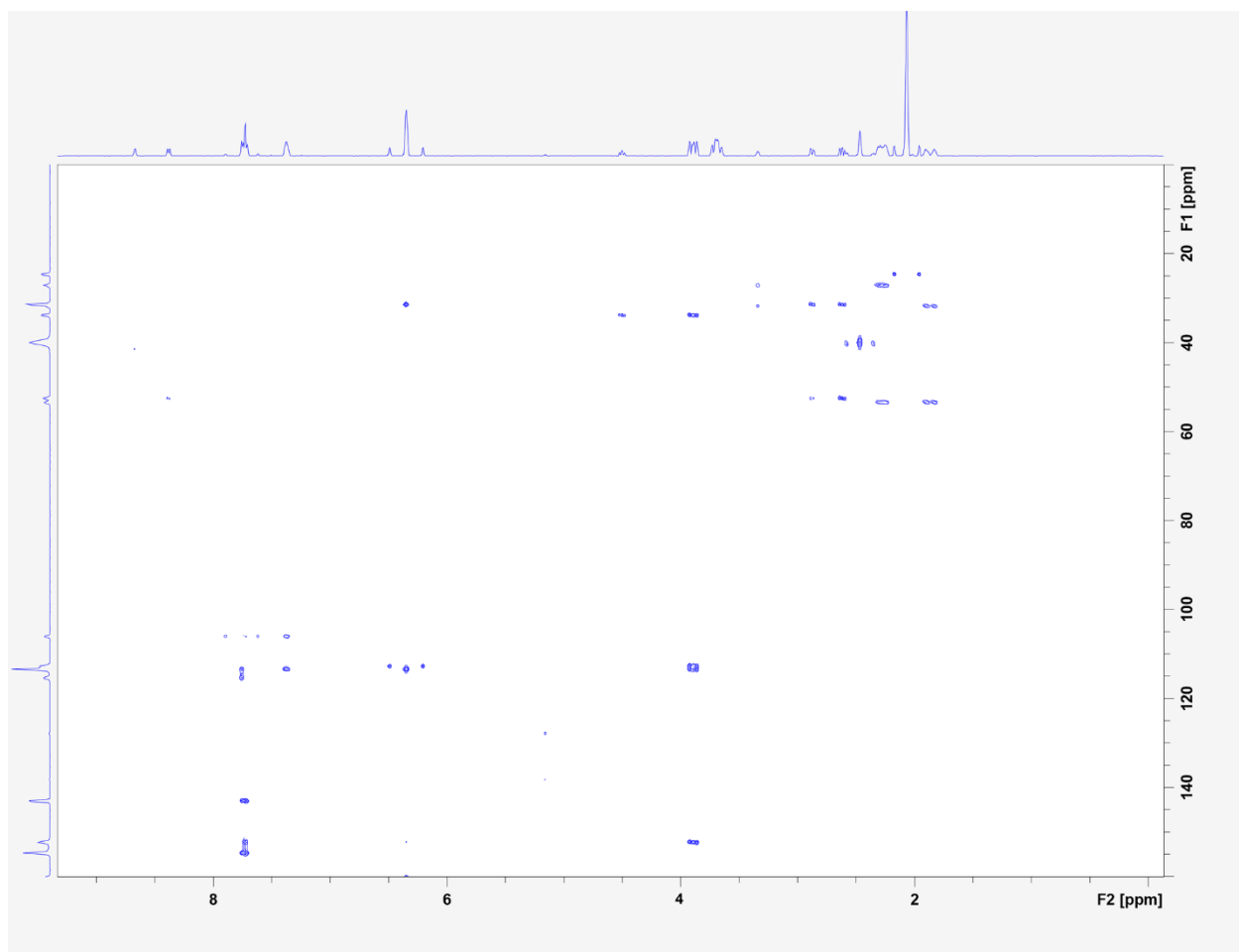


Figure S15. HMBC of 1-GSH.

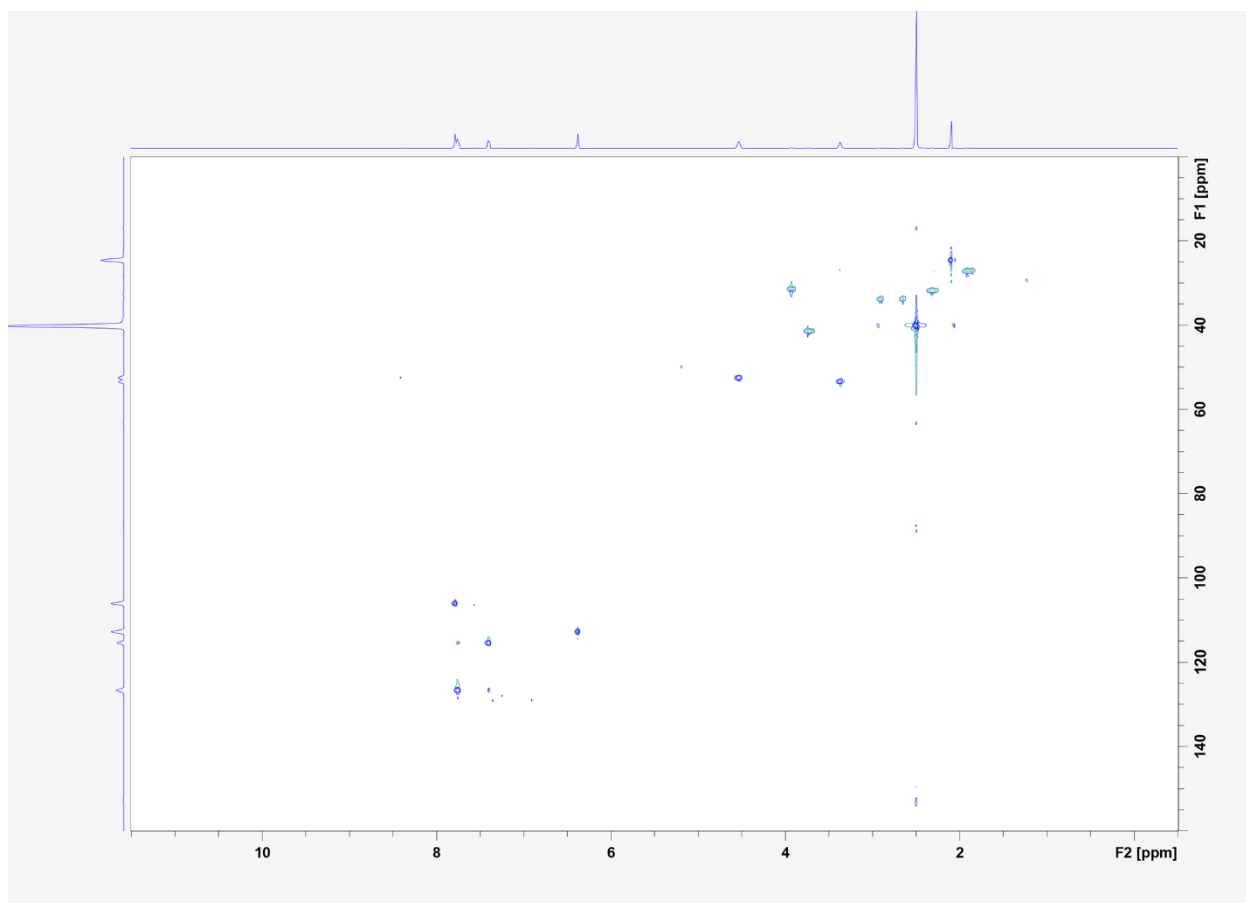
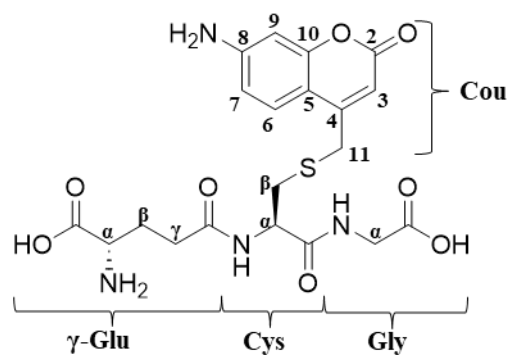


Figure S16. HSQC of 1-GSH.

Synthesis of 1a-GSH



1a-GSH was synthesized according to general procedure A using coumarin derivative **1a** (0.4 mmol, 83 mg).

The crude material was purified by preparative RP-HPLC using a gradient of 4%/min (5-95% acetonitrile in MilliQ water with 0.1% formic acid), yielding **1a-GSH** as a slightly yellow powder (27 mg, 14%; 481.01 amu). ^1H NMR, ^{13}C NMR, COSY, HMBC and HSQC were measured (**Figure S17-S22**).

^1H NMR (600 MHz, DMSO) δ 8.69 (t, GlyHN), 8.44 – 8.37 (d, CysHN), 7.45 (d, $J = 8.7$ Hz, CouH9), 6.55 (dd, $J = 8.7, 2.2$ Hz, CouH7), 6.40 (d, $J = 2.2$ Hz, CouH6), 6.04 (s, CouH3), 4.54 (dd, $J = 8.9, 4.3$ Hz, CysH α), 3.85 – 3.79 (s, CouH11), 3.75 – 3.70 (s, GlyH α), 3.61 (GluH α)*, 2.89 (dd, $J = 13.6, 4.4$ Hz, CysH β 1), 2.65 (d, $J = 12.2, 9.0, 2.7$ Hz, CysH β 2), 2.32 (t, $J = 8.8$ Hz, GluH γ), 1.92 (q, $J = 17.2, 8.0$ Hz, GluH β).

Note: *GluH α (3.61) overlays with solvent peak. Assigned according to ^1H - ^1H COSY/HSQC.

^{13}C NMR (151 MHz, DMSO) δ 171.4 (GluCOH), 170.9 (Cys=O), 170.6 (GluC=O), 170.5 (GlyC=O), 160.6 (CouC2), 156.0 (CouC8), 153.1 (CouC10), 152.7 (CouC4), 126.5 (CouC9), 111.1 (CouC7), 107.5 (CouC3) 107.0 (CouC4), 98.7 (CouC6), 52.4 (GluC α), 52.0 (CysC α), 40.9 (GlyC α), 33.7 (CysC β), 31.2 (GluC γ)*, 31.0 (CouC11)*, 26.4 (GluC β).

Note: *Differentiation is not possible between CouC11 (31.0) and GluC γ (31.2); assignment could be vice versa.

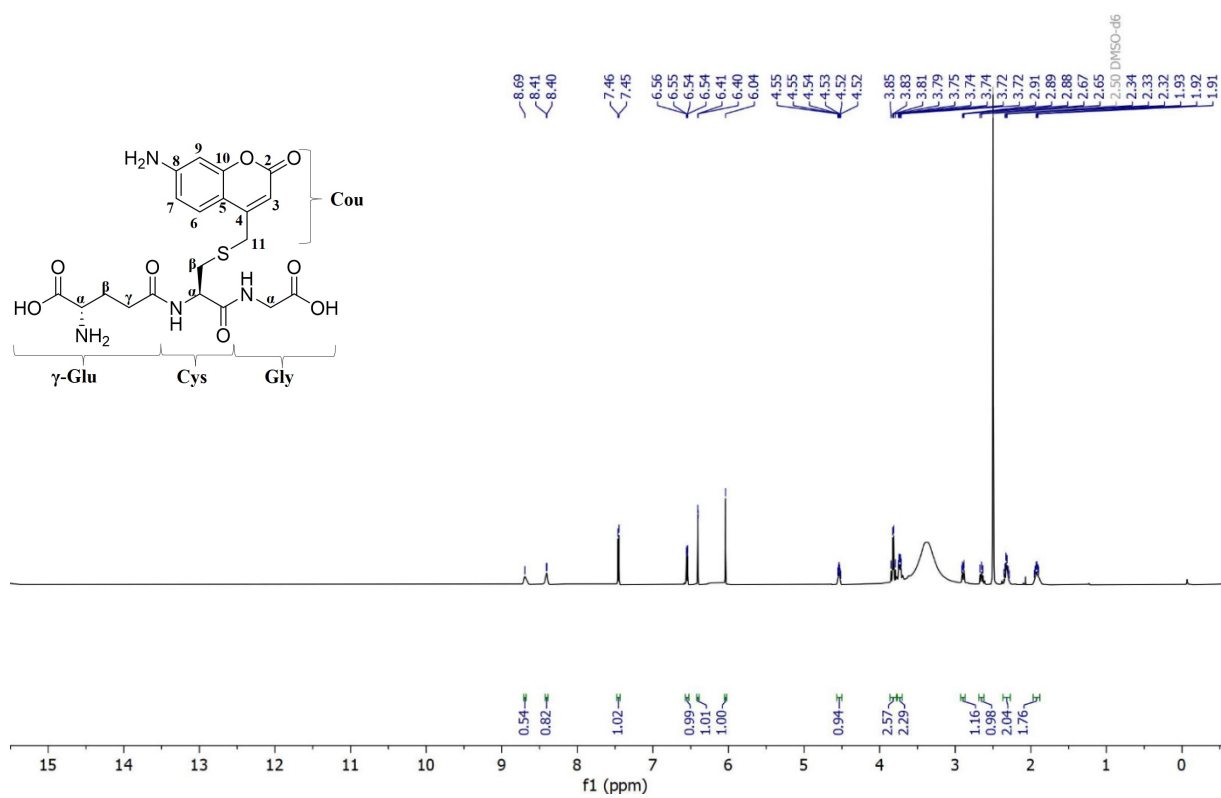


Figure S17. ¹H-NMR of 1a-GSH.

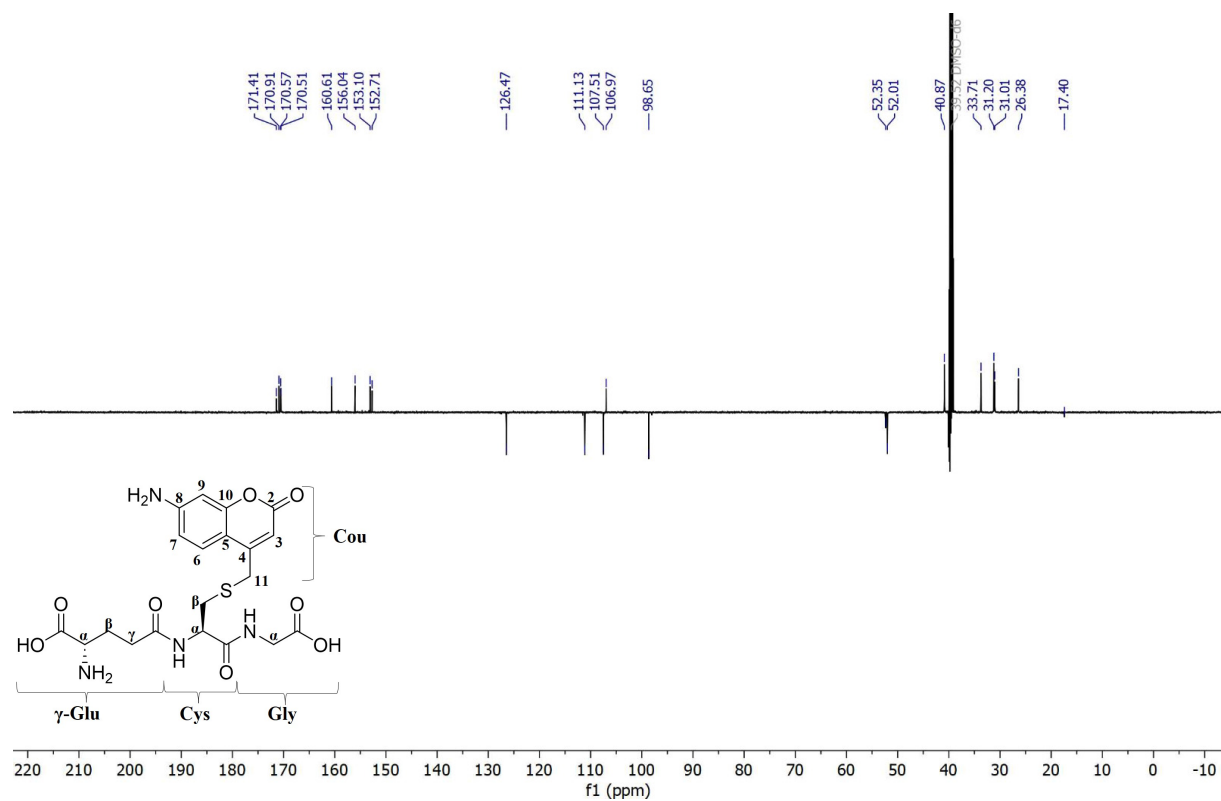


Figure S18. ^{13}C -NMR of 1a-GSH.



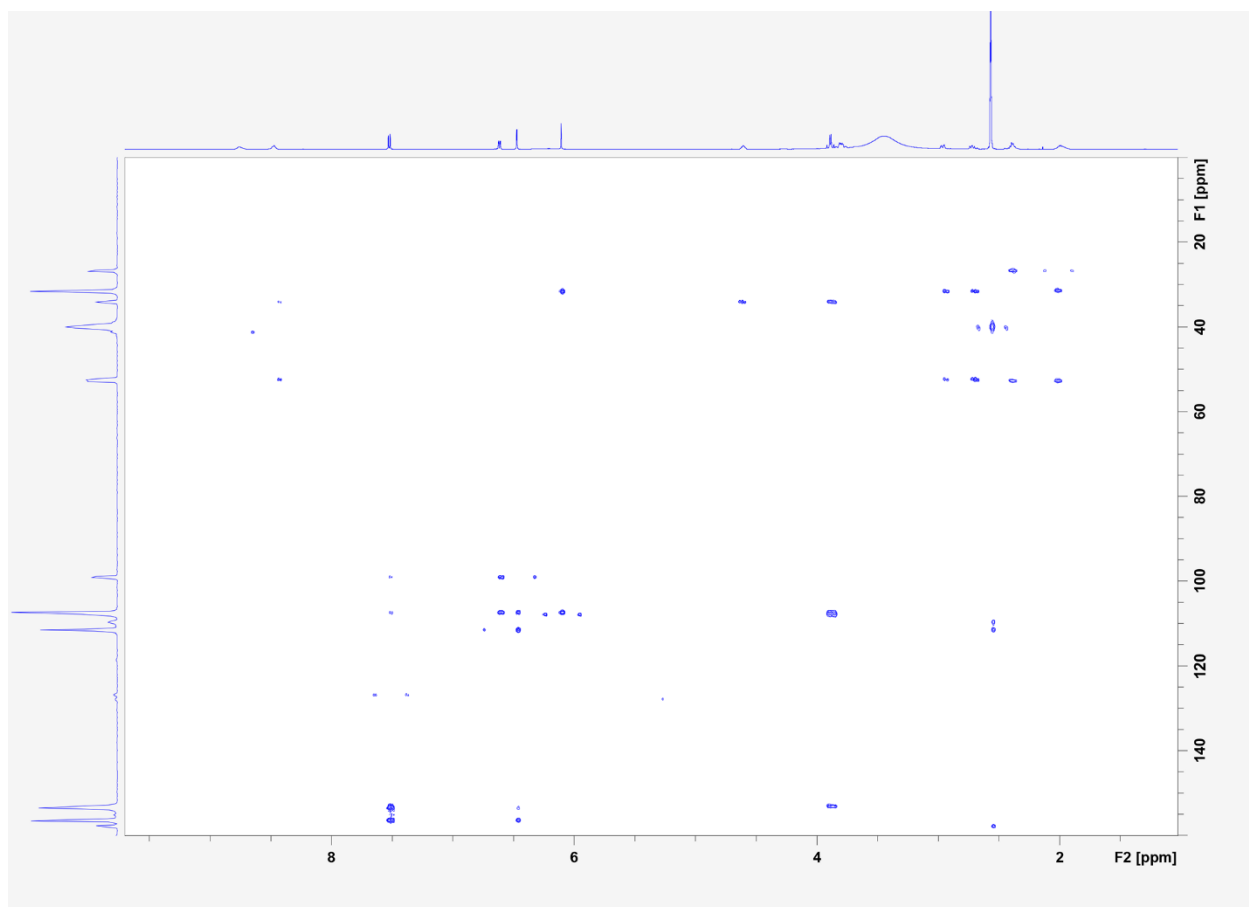


Figure S20. HMBC of **1a-GSH**.

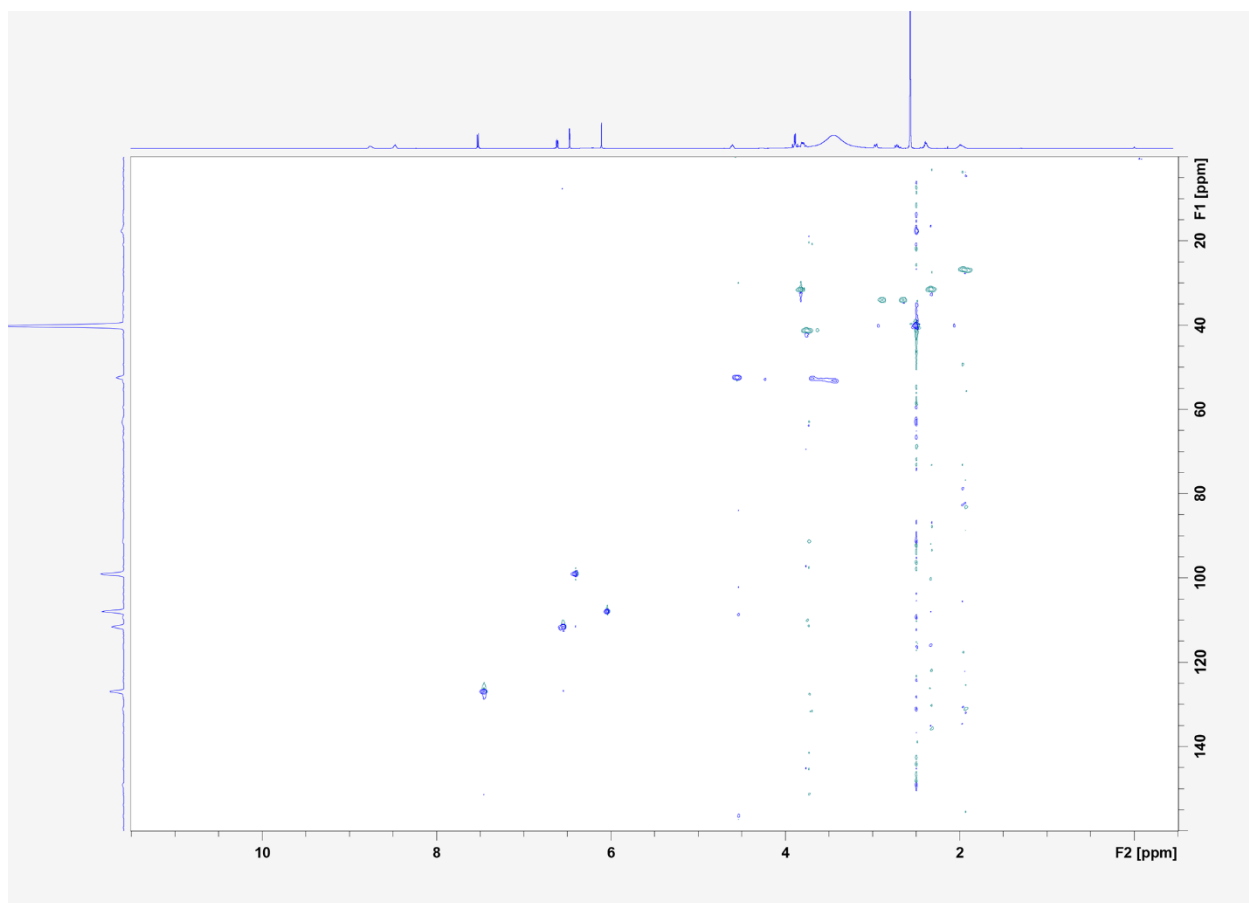


Figure S21. HSQC of **1a-GSH**.

Protein Sequences

***SaAmd* with C-terminal 6xHis-tag**

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NVPVSLADLQSVNPVYGRTSNPHDLSRVPGGSSGGAAAALASGMVPLEYGSDIGGSIRV
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QSDYMRMLGIAIARGLAPQGGTPATLIQWFDLLDDQARNKRAWVRLFEEFDVVFAPVL
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VIAAPHRDHTAIAVARLVDGAIGLELEHHHHHH

***PdAmd* with C-terminal 6xHis-tag**

MSLSNDLSYMSVAELSSKILRKQLSPVDVIEAVITRIKERNESLNAFVHVDVFEGAKEMAK
KAEAEIMSGAEVGIMHGIPSAIKDLDFDKPGWPSTFGGIRALKHNITDFYCPYAERMEKS
GAILVGKTNSPVMGFRGTCDNYLFGPSRNPFLNRKNTGGSSGGSAAAVADGLLPIAEGT
DGGGSIRIPAAWCGLYGYKASFGRVPSVARPNAFGGINPFLFEGTLTRNVEDAAIGLTAL
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KGYKMNVLDLNRDQHLRTEIYDAIQNILNKYDLIITPTVASLPVDNTNDGNTKGPTQING
VEVDPLIGWCLTYFTNFTGHPSASIPAGLSKENLPIMQIIGRRYADTDVLAASSVFERM
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TrAmd with C-terminal 6xHis-tag

MTMHEELAYMSATELALHIRRRQLSPVEVVDAFITRIEQRNPSLNAFVYFGFDEARQRA
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ERAGAIVLGKTNSPVMGFRGTCDNYLFGPSRNPFDLRRNTGGSSGGSAAAVADGLVPF
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PSQVNGESVDPLIGWCLTYPVNFSGHPAASIPAGLSASGLPVGLQIIGRRYADADVLTA
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GST

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LKVDFLSKLPEMLKMFEDRLCHKTYLNGDHVTHPDFMLYDALDVVLHMDPMCLDAFP
KLVCFKKRIEAI PQIDKYLKSSKYIAWPLQG WQATFGGGDHPPKSDHLVPRH

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